BioSense-Edge

Precision Bioprocess Environment & Telemetry Monitoring

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CORE PLATFORM
STM32H743ZI + ESP32

REPOSITORY
GitHub ↗

Abstract

BioSense-Edge is a dual-microcontroller edge telemetry system engineered to eliminate micro-environmental parameter drift in bioprocess and fermentation batches. A deterministic STM32H743ZI core executes real-time closed-loop PID correction across pH, dissolved oxygen, temperature, and optical density (OD600) channels, while an ESP32 co-processor independently manages Wi-Fi/MQTT connec-tivity with automatic SX1276 LoRa failover. Every analog sensing channel is galvanically isolated via ISO1540 isolators to guarantee signal integrity in a noisy bioreactor electrical environment. At a total bill of materials of ₹23,000 ($280 USD), the system delivers a 10×–36× cost reduction relative to imported legacy monitoring hardware priced between $3,000 and $10,000, while remaining fully open-source and reproducible.

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1 Problem Statement

1.1 Micro-Environmental Drift as a Batch-Critical Failure Mode

High-value biopharmaceutical, enzyme, and fermentation processes operate within narrow tolerance bands for pH, dissolved oxygen (DO), temperature, and biomass density (OD600). Undetected drift in any single parameter — often on the order of minutes to hours — can irreversibly shift metabolic pathways, denature product yield, or trigger contamination risk, voiding an entire production batch.

Conventional laboratory practice relies on periodic manual sampling or coarse analog controllers with no closed-loop correction, no black-box audit trail, and no redundant telemetry path. This leaves a substantial blind-spot window between measurement events during which drift is entirely unmonitored.

1.2 The Cost Barrier to Precision Monitoring

Commercial, import-grade bioprocess telemetry and control systems capable of closed-loop correction are priced well outside the reach of most academic labs, startups, and mid-scale manufacturers operating in cost-sensitive markets.

Import Cost Barrier

Legacy imported bioprocess controllers typically range from **$3,000 to $10,000 USD** (approximately **₹2.5,00,000 to ₹8,30,000 INR**) per unit — a cost structure that effectively excludes small-scale fermentation R&D units, university labs, and early-stage agri-biotech ventures from real-time precision monitoring.

1.3 Target Deployment Context

- **Biopharma QC laboratories** requiring audit-grade environmental logs
- **Fermentation R&D units** iterating on strain and media conditions
- **Vaccine and enzyme production facilities** operating at pilot scale
- **Agri-biotech startups** without capital budget for imported instrumentation

BioSense-Edge is architected specifically to close this gap: deterministic, closed-loop, locally-sourced, and reproducible at a fraction of legacy system cost.

2 System Architecture & Dual-MCU Offloading Strategy

2.1 Design Rationale

BioSense-Edge deliberately partitions compute across two independent microcontrollers rather than a single unified processor. This separation exists to guarantee that the real-time control loop is never blocked, delayed, or jittered by network-layer non-determinism such as Wi-Fi negotiation, MQTT broker latency, or DNS resolution stalls.

STM32H7 — Main MCU	ESP32 — Network Co-Processor
Deterministic real-time domain @ 480 MHz	Connectivity domain, fully offloaded
High-speed synchronized ADC/SPI sampling	Wi-Fi / MQTT stack — primary telemetry uplink
Closed-loop PID correction engine	SX1276 LoRa transceiver — automatic failover
Local SD card black-box logger plus ETH PHY	Setpoint/command relay back to PID loop

2.2 Inter-MCU Communication

The two domains are bridged over a **UART/SPI link at 3.3V logic levels**, carrying telemetry upstream (sensor readings, system state) and commands downstream (operator setpoint changes issued from the cloud dashboard). This link is bidirectional but non-blocking: the STM32H7 continues its control loop irrespective of link state.

```
[ Sensing + Analog FE ] -> [ STM32H743ZI: ADC/SPI -> PID -> SD Logger ]
                                    |     UART / SPI (3.3V)
                                    v
                       [ ESP32: Wi-Fi/MQTT <-> SX1276 LoRa ]
                                    |
                                    v
                    [ Cloud Dashboard / InfluxDB / Telegram Alerts ]
```

2.3 Why This Matters for Reliability

- A dropped Wi-Fi packet **never** delays a PID correction cycle
- Pump/heater actuation timing remains hard-real-time regardless of network state
- Network co-processor can restart or re-associate independently without halting control

3 Hardware & Signal Isolation Chain

3.1 Sensing Channels

Parameter	Sensor / Front-End	Interface
pH	Glass electrode probe + Atlas Scientific EZO-pH circuit	I²C
Dissolved oxygen	Galvanic/optical DO probe + Atlas Scientific EZO-DO circuit	I²C
Temperature	PT100 RTD (3-wire) + MAX31865 driver	SPI
Optical density	660 nm photodiode pair + transimpedance amp	SPI ADC

3.2 Galvanic Isolation Strategy

Bioreactor electrical environments are electrically hostile: peristaltic pump motors, solid-state relay switching, and wet-probe grounding create ground loops, ESD events, and common-mode noise that can silently corrupt low-level analog sensor signals.

Isolation Barrier

Every analog front-end channel is coupled to the STM32H7's I²C/SPI bus through **ISO1540 galvanic isola-tors**, rated for isolation withstand well in excess of typical bioreactor common-mode transients. This ensures sensor faults, ground loops, or actuator switching noise on the "wet" analog side can never propagate into or corrupt the digital MCU domain.

3.3 Actuation Layer

- **Solid-state relay (SSR)** — zero-cross switched heating element control
- **Peristaltic dosing pump A** — acid dosing, stepper-driven
- **Peristaltic dosing pump B** — base dosing, stepper-driven

All actuator drive signals originate from the STM32H7 PID engine via PWM/GPIO, downstream of the same isolation boundary that protects the sensing chain —— actuator switching noise is prevented from re-entering the analog measurement path.

4 Software Methodology & Network Watchdog Logic

4.1 Closed-Loop Control Sequence

1. **Sense** — All four channels sampled synchronously via isolated ADC/SPI, decoupled from network timing
2. **Compute** — STM32H7 evaluates independent PID loops per channel against configured setpoints
3. **Actuate** — Correction signals drive the SSR heater and dosing pumps in real time
4. **Log** —– Every sample and correction event is written to the local SD black-box log, independent of network availability

4.2 Network Watchdog & Fail-Safe Routing

The ESP32 continuously monitors Wi-Fi/MQTT link health via periodic keepalive and broker acknowledgement checks. On detecting sustained link failure, it autonomously reroutes outbound telemetry over the SX1276 LoRa transceiver without operator intervention.

```
watchdog_loop():
    if mqtt.connected() and wifi.rssi() > THRESHOLD: publish(telemetry,
        via = WIFI_MQTT) failover_state = PRIMARY
    else:
        publish(telemetry, via = LORA_SX1276) failover_state = FALLBACK
        attempt_wifi_reconnect_async()
```

Key properties of this watchdog design:

- **Non-blocking** — reconnection attempts run asynchronously and never stall telemetry publishing
- **Stateless recovery** — the STM32H7 control loop is entirely unaware of which uplink is active
- **Zero manual intervention** — operators retain full visibility through outage and recovery

4.3 Data Integrity Guarantee

Because the SD black-box logger operates independently of both the ESP32 and its uplink state, the system guarantees **zero data loss** at the edge even during extended network outages. Buffered records are available for later synchronization to the cloud historian via the Ethernet TCP bridge described in Section 7.

5 Bill of Materials (BOM) & Economic Viability

5.1 Detailed Component Cost Breakdown

Component	Function	Qty	Est. Cost (₹)
STM32H7	Main deterministic MCU	1	2000
ESP32	Network co-processor	1	650
SX1276 LoRa module	868 MHz failover radio	1	850
Atlas Scientific EZO-pH circuit	pH signal conditioning	1	3,400
Atlas Scientific EZO-DO circuit	DO signal conditioning	1	3,600
PT100 RTD probe	Temperature sensing	1	450
MAX31865 RTD amplifier	RTD linearization/driver	2	1000
OD600 photodiode assembly	Optical density sensing	1	900
ISO1540 galvanic isolators	Signal isolation barrier	2	560
Solid-state relay (SSR)	Heating element control	2	1000
Peristaltic dosing pumps	Acid/base dosing	2	2,600
microSD module + card	Black-box logging	1	400
Enclosure, wiring, PCB, misc.	Mechanical + interconnect	——	3,810
Total			**~ ₹23,000**

5.2 Cost Comparison

₹23,000
BioSense-Edge total BOM ($280 USD)

₹2.5L – 8.3L
Legacy imported system ($3,000-$10,000)

10×–36×
Cost reduction achieved

5.3 Feasibility & Localization

Every listed component is available off-the-shelf through Indian and global electronics distributors at the time of writing —— no custom fabrication, application-specific integrated circuits, or import-only parts are required. This directly supports rapid replication, local manufacturing, and educational reproducibility.

6 Verification, Simulation & Repository Deliverables

6.1 Verification Approach

- **Bench-level signal verification** — isolated I^2C/SPI bus integrity confirmed under simulated pump/SSR switching noise
- **PID loop tuning** — closed-loop step-response validated in a controlled water-bath rig prior to live culture deployment
- **Failover testing** — Wi-Fi link forcibly interrupted to confirm sub-second LoRa fallback and telemetry continuity
- **Data integrity testing** — SD black-box log cross-checked against cloud-synchronized records post-outage

6.2 Repository & Reproducibility

Open-Source Deliverable

Full firmware source (STM32H7 + ESP32), schematic references, bill of materials, and build documentation are published and version-controlled at:
github.com/GJVIJAYG/BioSense-Edge

6.3 Summary

BioSense-Edge demonstrates that deterministic, closed-loop bioprocess monitoring — previously acces-sible only through costly imported instrumentation — can be achieved with locally-sourced components, an open architecture, and rigorous signal isolation practice, at roughly one-tenth to one-thirty-sixth the cost of legacy alternatives. While using custom Application processors like I.MX 9 from manufacturer like NXP an dedicated SBC can be built and customized linux can be used to control the real time stm32 via ETH . Due to the presence of eIQ neuron accelerator present in I.MX 9 rated @ 2TFLOPS local Ai stuff could be processed on board locally .